LINEAR ALGEBRA SERIES

The Backbone of Modern Computing

Linear algebra powers the technology shaping our world — from machine learning to graphics and robotics. This deck explores its role in data representation, computation, and beyond.

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MACHINE LEARNING & DATA SCIENCE

Why Linear Algebra Matters in ML & Data Science

Data Representation

Vectors and matrices represent complex data structures inside machine learning models.

Computational Efficiency

Linear algebra enables fast, large-scale computation through vectorized operations.

Neural Networks

Matrix operations power gradient calculations and transformations in deep learning.

Key Applications Across Industries

01

🖥️ Computer Graphics

3D object transformations rely on matrix operations for rotation and scaling.

02

🤖 Robotics

Vectors and matrices guide movement and transformations in robotic systems.

03

📶 Signal Processing

Fourier and wavelet transforms utilize linear algebra for data analysis.

Fundamental Concepts in Linear Algebra

Vectors & Vector Spaces

1

Represent multi-dimensional data points and features.

Matrices

2

Transform data across a huge range of applications.

Eigenvalues & Eigenvectors

3

Find principal components in data analysis.

Matrix Factorization

4

Used in data compression and signal processing.

Real-World Examples in Data Science



Face Recognition

Eigenfaces use linear algebra to represent facial features as basis vectors.



Recommendation Systems

Matrix factorization predicts user preferences in platforms like Netflix and Amazon.



Image Processing

CNNs apply matrix operations to detect edges and patterns in images.